Teacher Information, Course Description and Lecture-wise Plan

of

Database Systems (CSC270)

under

Pilot Project:
Uniform Teaching, Assessment and Quality Assurance Across All The Campuses

by

Faculty of Information Science and Technology
COMSATS University Islamabad

A: Teacher Information

Course Name: Database Systems	Credit Hours: 3+1
Course Code: CSC270	Pre-requisite:
Teacher Name:	Teacher E-mail:
Program: BS (Computer Science)	Semester: Fall 2026
Department: Computer Science	Campus:

B: Course Contents

This course introduces the fundamental concepts of database systems. Topics include: Introduction to Databases & Information Systems; Evolution of Database Systems; Components; Architecture; Functions; Relational Model; Relational Algebra; Relational Calculus; Data Modeling; Relational Data Model; Relational Algebra & Calculus; Integrity Constraints; Conceptual Models; Entity-Relationship (E-R) Model; Enhanced E-R Model; Mapping Conceptual Schema to Relational Schema; Functional Dependency & Normalization; Structured Query Language (SQL); Views; Materialized Views; Non-Relational/No SQL Databases; MongoDB as NoSQL Database; Document Model; and Transaction Management.

Recommended Books:

Text Books:

1. Database systems: A Practical Approach to Design, Implementation, and Management, Thomas Connolly, Carolyn Begg, Pearson, 2015.
2. MongoDB: The Definitive Guide, Shannon Bradshaw, Eoin Brazil, Kristina Chodorow, O'Reilly Media, 2019.

Reference Books:

3. Fundamentals of Database Systems, Elmasri, R, Navathe, Pearson, 2016.
4. Database System Concepts, Silberschatz, Korth, Sudarshan, McGraw Hill, 2019.

C: Lecture-wise Plan

Week #	Lecture #	Topic (Chapter/Topics Covered)	Practice/ Evaluation	Remarks
1	1 (Date)	***Introduction to Databases:*** • *Introduction to Databases* • *Information Systems* • *Database Applications* • *Evolution of DBMS* • *File-Based Systems* • *Database Approach* • *Advantages of DBMS* • *Disadvantages of DBMS*	*Exercises as given in the books / Real-world case studies*	*Connolly : Ch1*
	2 (Date)	***Database Approach:*** • *Data & Metadata* • *Database Environment* • *Approaches & Roles* • *Components* • *Functions* • *Data Dependences*	*Exercises as given in the books / Real-world case studies*	*Connolly : Ch1*
2	3 (Date)	***Database Languages & Functions:*** • *Data Definition Language (DDL)* • *Data Manipulation Language (DML)* • *Data Control & Security* • *DBMS Functions* • *Database Utilities* • *Database Administration Functions*	*Exercises as given in the books / Real-world case studies*	*Connolly : Ch2*
	4 (Date)	***Database Architectures & Data Independence*** • *DBMS Architecture* • *Two-Tier Client–Server Architecture* • *Three-Tier Client–Server Architecture* • *Three-Schema Architecture* • *External Level* • *Conceptual Level* • *Internal Level* • *External–Conceptual Mapping* • *Conceptual–Internal Mapping* • *Data Independence* • *Logical Data Independence* • *Physical Data Independence*	***Quiz # 1***	*Connolly : Ch2, Ch3*
3	5 (Date)	***Relational Model:*** • *Overview of the Relational Model* • *Notion of a Relation* • *Mathematical Relations* • *Database Relations*	*Exercises as given in the books*	*Connolly : Ch4*

		• *Relation Properties* • *Attributes and Domains* • *Relational Keys* • *Candidate, Primary & Foreign Keys* • *Relational Database Schema* • *Representing Relational Database Schemas*		
	6 (Date)	***Integrity Constraints:*** • *Domain Constraints* • *Entity Integrity* • *Referential Integrity* • *General Constraints* • *Integrity Constraints in Relational Databases* • *Enforcement of Integrity Constraints*	*Exercises as given in the books*	*Connolly : Ch4*
4	7 (Date)	***Relational Algebra:*** • *Introduction to Relational Algebra* • *Unary Relational Operations* ○ *SELECT* ○ *PROJECT* • *Set Operations* ○ *UNION* ○ *INTERSECTION* ○ *DIFFERENCE* ○ *CARTESIAN PRODUCT*	*Exercises as given in the books + Assignment No 1*	*Connolly : Ch5*
	8	***Join & Division Operations:*** • *Join Operations* • *Theta Join* • *Equijoin* • *Natural Join* • *Outer Joins* • *Division Operation* • *Examples of Join & Division Operations*	***Quiz # 2***	*Connolly : Ch5*
5	9 (Date)	***Aggregation & Grouping:*** • *Aggregate Functions* • *Grouping Operations* • *Grouping with Aggregate Functions* • *Examples of Aggregation & Grouping*	*Exercises as given in the books*	*Connolly : Ch5*
	10 (Date)	***Relational Calculus:*** • *Introduction to Relational Calculus* • *Tuple Relational Calculus (TRC)* • *Tuple Variables* • *Domain Relational Calculus (DRC)* • *Domain Variables* • *Tuple vs. Domain Relational Calculus*	*Exercises as given in the books*	*Connolly : Ch5*

6	11 (Date)	***Data Modeling & Database Design:*** • *Data Modeling* • *Conceptual Data Models* • *Logical Data Models* • *Relational Database Design* • *Mapping Conceptual Model to Logical Model* • *Mapping Logical Model to Relational Schema* • *Relational Database Design Considerations*	***Quiz # 3***	*Connolly : Ch10*
	12 (Date)	***Entity-Relationship (ER) Model:*** • *Introduction to the ER Model* • *Entity Types* • *Relationship Types* • *Attributes* • *Strong & Weak Entity Types* • *Attributes on Relationships* • *Structural Constraints* • *Cardinality Constraints* • *Participation Constraints*	*Exercises as given in the books / Real-world case studies*	*Connolly : Ch12*
7	13 (Date)	***ER Modeling Issues/Problems:*** • *Identifying Keys* • *Naming Conventions* • *Handling Complex Relationships* • *Structural Constraints* • *Avoiding Redundancy* • *Modeling Generalization & Specialization* • *Common ER Modeling Errors*	*Exercises as given in the books / Real-world case studies*	*Connolly : Ch12*
	14 (Date)	***Enhanced ER (EER) Model:*** • *Enhanced ER Model* • *Specialization* • *Generalization* • *Superclasses & Subclasses* • *Inheritance* • *Constraints on Specialization/Generalization* • *Aggregation* • *Composition* • *Categorization / Union Types*	*Exercises as given in the books / Real-world case studies + Assignment No 2*	*Connolly : Ch13*
8	15 (Date)	***EER Modeling Case Study:*** • *Case Study Description* • *Identification of Entity Types* • *Identification of Relationships* • *Identification of Attributes* • *Specialization & Generalization* • *Superclass/Subclass Relationships* • *Inheritance* • *Aggregation & Composition*	*Exercises as given in the books / Real-world case studies*	*Connolly : Ch13*

		• *EER Diagram Development* • *Validation of EER Model*		
	16	***Mapping EER to Relational Schema:*** • *Mapping EER Model to Relations* • *Mapping Strong Entity Types* • *Mapping Weak Entity Types* • *Mapping 1:1 Relationships* • *Mapping 1:N Relationships* • *Mapping M:N Relationships* • *Mapping Multivalued Attributes* • *Mapping Specialization & Generalization* • *Mapping Categories / Union Types* • *Mapping Aggregation* • *Mapping EER Constraints to Relational Schema*	*Exercises as given in the books / Real-world case studies*	*Connolly : Ch17*
9	17 (Date)	*Midterm Examination*	***Midterm (25%)***	
	18 (Date)	***Normalization:*** • *Purpose of Normalization* • *Normalization in Database Design* • *Data Redundancy* • *Update Anomalies* • *Functional Dependencies* • *Functional Dependency and Normalization* • *Decomposition of Relations* • *Lossless-Join Property* • *Dependency-Preservation Property* • *Normalization Process*	*Exercises as given in the books / Real-world case studies*	*Connolly : Ch14*
10	19 (Date)	***Normalization Process:*** • *First Normal Form (1NF)* • *Second Normal Form (2NF)* • *Third Normal Form (3NF)* • *Conversion to 1NF* • *Conversion to 2NF* • *Conversion to 3NF* • *Examples of Normalization* • *Testing for Normal Forms*	*Exercises as given in the books / Real-world case studies*	*Connolly : Ch14*
	20 (Date)	***Advanced Normalization:*** • *Functional Dependency Inference Rules* • *Closure of Functional Dependencies* • *Attribute Closure* • *Boyce-Codd Normal Form (BCNF)* • *BCNF vs. 3NF*	*Exercises as given in the books / Real-world case studies*	*Connolly : Ch15 & Elmasri Chapter No 15*

11	21 (Date)	***BCNF Schema Design by Decomposition Algorithm:*** • *BCNF Decomposition Algorithm* • *Identifying BCNF Violations* • *Decomposing Relations* • *Lossless-Join Decomposition* • *Testing Relations for BCNF* • *Iterative BCNF Decomposition* • *Dependency Preservation* • *Worked Examples of BCNF*	*Exercises as given in the books + Assignment No 3*	*Connolly : Ch15 & Elmasri Chapter No 15*
	22 (Date)	***Multi-Valued Dependencies and Fourth Normal Form:*** • *Multi-Valued Dependencies (MVDs)* • *Trivial & Non-Trivial MVDs* • *Properties of MVDs* • *Inference Rules for MVDs* • *Fourth Normal Form (4NF)* • *4NF Decomposition* • *Lossless-Join Property in 4NF* • *Examples of 4NF Decomposition*	*Exercises as given in the books / Real-world case studies*	*Connolly : Ch15*
12	23 (Date)	***Views & Materialized Views:*** • *Introduction to Views* • *Creating Views* • *Querying Views* • *Updating Views* • *Advantages & Disadvantages of Views* • *Materialized Views* • *Creating & Refreshing Materialized Views* • *Advantages & Disadvantages of Materialized Views*	***Quiz # 4***	*Connolly : Ch7*
	24 (Date)	***Database Authorization:*** • *Database Security & Authorization* • *Database Users & Privileges* • *Access Control* • *Granting Privileges* • *Revoking Privileges* • *Roles & Role-Based Access Control* • *Authorization Rules* • *Views for Security* • *SQL Authorization Commands* • *Security Considerations*	*Exercises as given in the books*	*Connolly : Ch20*
13	25 (Date)	***Indexing:*** • *Introduction to Indexing*	*Exercises*	*Connolly :*

		• *Purpose of Indexes* • *Primary Indexes* • *Secondary Indexes* • *Dense & Sparse Indexes* • *Single-Level Indexes* • *Multi-Level Indexes* • *Index Selection & Performance* • *Advantages & Disadvantages of Indexing*	*as given in the books*	*Ch18*
	26 (Date)	***Introduction to NoSQL/Non-Relational Databases:*** • *Introduction to NoSQL Databases* • *Need for NoSQL* • *Characteristics of NoSQL Databases* • *Relational vs. NoSQL Databases* • *Types of NoSQL Databases* • *NoSQL Data Models* • *Advantages & Limitations* • *NoSQL Use Cases* • *CAP Theorem Overview*	*Exercises as given in the books*	*Shannon : Ch1*
14	27 (Date)	***MongoDB & Document Model:*** • *Introduction to MongoDB* • *MongoDB Architecture* • *Document-Oriented Data Model* • *Documents and Collections* • *BSON Data Format* • *MongoDB Data Types* • *Embedded Documents & Arrays* • *MongoDB CRUD Operations* • *MongoDB Querying* • *MongoDB Use Cases*	*Exercises as given in the books / Real-world case studies*	*Shannon : Ch2*
	28 (Date)	***MongoDB Sharding & Replication:*** • *MongoDB Replication* • *Primary & Secondary Nodes* • *Automatic Failover* • *Read & Write Operations in Replication* • *MongoDB Sharding* • *Sharded Cluster Architecture* • *Query Routing & mongos* • *Balancing & Data Distribution* • *Sharding vs. Replication*	*Exercises as given in the books / Real-world case studies + Assignment No 4*	*Shannon : Ch9 and Ch13*
15	29 (Date)	***Database Transactions & Properties:*** • *Introduction to Database Transactions*	*Exercises*	*Connolly :*

		• *Transaction Processing* • *Transaction States* • *ACID Properties* • *Atomicity* • *Consistency* • *Isolation* • *Durability* • *Importance of Transactions* • *Transaction Management in DBMS*	*as given in the books*	*Ch22*
	30 (Date)	***Transaction Management:*** • *Concurrency Control* • *Serial & Concurrent Schedules* • *Lock-Based Concurrency Control* • *Two-Phase Locking (2PL)* • *Deadlocks* • *Isolation Levels* • *Read Phenomena* • *Transaction Isolation & Consistency*	*Exercises as given in the books*	*Connolly : Ch22*
16	31 (Date)	***Failure and Recovery in Transaction Management:*** • *Types of Transaction Failures* • *System & Media Failures* • *Recovery Concepts* • *Log-Based Recovery* • *Undo & Redo* • *Checkpointing* • *Rollback & Rollforward* • *Database Recovery Procedures*	*Exercises as given in the books*	*Connolly : Ch22*
	32 (Date)	• *Review*	*Group Presentations (optional)*	
17	(Date)		***Final (50%)***	

Note: All Assignments/Quizes weightage is 25%